

THE GLP-1 LIFESTYLE SERIES

GLP-1 and Strength Training

Building Muscle While on Ozempic

Disclaimer

This book is for educational and informational purposes only. It is not medical advice and does not constitute a recommendation to start, stop, or change any medication or treatment. GLP-1 receptor agonists are prescription medications that should only be used under the supervision of a qualified healthcare provider.

The nutritional guidance, exercise information, and lifestyle strategies in this book are general wellness information and are not tailored to your individual medical situation. Always consult your doctor, registered dietitian, or other qualified health professional before making changes to your diet, exercise routine, or medication management.

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Why Strength Training Is Non-Negotiable on GLP-1

GLP-1 medications create a calorie deficit. Calorie deficits, without resistance training, result in the loss of muscle alongside fat. Strength training is the primary tool for ensuring the weight you lose is the weight you actually want to lose.

The Research on Muscle Loss with GLP-1

Multiple clinical trials and real-world studies have documented significant lean mass loss alongside fat loss on GLP-1 medications. Without countermeasures, 25–40% of the weight lost can be lean tissue. Over a year of treatment losing 30 pounds, that could mean 8–12 pounds of muscle lost alongside 18–22 pounds of fat.

The long-term metabolic consequences of this are significant. Each pound of muscle burns approximately 6–10 calories per day at rest. Losing 10 pounds of muscle reduces resting metabolic rate by 60–100 calories per day — a reduction that compounds over time and contributes to weight regain when medication stops.

What Resistance Training Does

Progressive resistance training provides a consistent anabolic stimulus — a signal to the body that muscle is being used and needed. Even in a calorie deficit, consistent resistance training dramatically reduces muscle loss and, in some cases (particularly in previously untrained individuals), can produce muscle gain alongside fat loss.

The combination of adequate protein intake and progressive resistance training is the evidence-based strategy for body recomposition on GLP-1. This is not a peripheral recommendation — it is the central intervention for ensuring that the medication's weight loss benefits translate into improved body composition rather than simply reduced scale weight.

"The scale is the wrong metric. Body composition is the right one."

Starting Strength Training on GLP-1

The first weeks on GLP-1 are not the time to start an aggressive new training programme. Here's how to sequence it correctly.

Phase One: Establish the Habit (Weeks 1–4)

In the first month on GLP-1, energy is reduced, nausea may be present, and the body is adapting to significant metabolic changes. The appropriate training goal in this phase is simply to establish a consistent movement habit without adding physiological stress.

Two sessions per week of light-to-moderate resistance training — bodyweight exercises or light weights, focusing on form rather than intensity — is appropriate. Keep sessions to thirty to forty minutes. Prioritise recovery. The goal is consistency, not intensity.

Phase Two: Progressive Loading (Months 2–3)

As the body adapts to the medication and side effects reduce, training intensity can increase. Move to two to

three sessions per week, progressively increasing the challenge through more weight, more repetitions, or more difficult exercise variations.

The key principle of progressive overload: the body adapts to the demands placed on it. Once a given level of challenge no longer produces adaptation, the challenge needs to increase. This is not about lifting maximum weight — it's about consistent, gradual progression over months.

The Essential Exercises

A functional strength programme doesn't require a gym. The movements that provide the greatest benefit for muscle preservation and metabolic health are: squats or leg press (lower body push), Romanian deadlifts or hip hinges (lower body pull/posterior chain), rows (upper body pull), and push-ups or chest press (upper body push).

These four movement patterns, trained twice weekly with progressive resistance, cover the major muscle groups and provide sufficient stimulus for lean mass preservation during GLP-1-assisted weight loss.

"Consistency beats intensity every time. Show up twice a week, progressively challenge yourself, and let time do the work."

Nutrition Timing Around Training

When you eat — and what you eat — around training sessions matters more when appetite is suppressed.

Pre-Workout Nutrition on GLP-1

Training on a completely empty stomach when appetite is suppressed is a common pattern on GLP-1 and a problematic one. Without pre-workout fuel, performance suffers, the workout stimulus is reduced, and muscle breakdown during the session is higher.

A small pre-workout meal or snack — thirty to sixty minutes before training, containing 20–30 grams of protein and some carbohydrate — significantly improves training performance and reduces muscle breakdown. If solid food is unappealing before training, a protein shake with a banana or piece of fruit is an effective alternative.

Post-Workout Protein

The post-workout period is when the muscle repair and synthesis process is most active. Consuming protein

within one to two hours of resistance training consistently improves the body's ability to build and preserve lean tissue from that training session.

Target 25–40 grams of protein in the post-workout meal or snack. Leucine is the key amino acid for post-workout muscle protein synthesis — found in highest concentrations in whey protein, eggs, and most animal proteins. A whey protein shake is practical and effective if appetite for solid food is limited post-workout.

"Feed the training. Even when you're not hungry, the muscle is asking."

Tracking Progress Beyond the Scale

Body composition changes on a well-executed GLP-1 plus training programme are often better than the scale suggests.

Measuring Body Composition

A DEXA scan is the gold standard for measuring lean mass and fat mass separately. Many gyms, hospitals, and wellness centres offer these for a modest fee, and a scan at the start and end of your programme gives you accurate data on whether your lean mass is being preserved.

Practical proxies if scanning isn't accessible: clothing fit (how do clothes that were tight in certain places now fit?), gym performance (are you lifting the same or more weight than when you started?), measurements at key sites (waist, hips, upper arm, thigh). Performance and measurements together tell you more than the scale alone.

The Long-Term Picture

The combination of GLP-1 medication and progressive strength training produces a result that neither achieves alone: significant fat loss with preserved or improved lean mass, resulting in a body composition that is metabolically healthier than weight loss without training produces. This combination is increasingly recognised in obesity medicine as the standard of care — medication alongside lifestyle intervention rather than medication alone.

"The goal is a stronger, leaner, more metabolically healthy body. The scale is one data point. Don't let it be the only one."
